

# Companion to Bird & de Moor, Algebra of Programming

## §6. Recursive Programs

### §6.1. Digits of a number

(6.1a)

$\text{val}^\circ = \text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val}^\circ) \text{snoc}$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(a) (b) (c) (d) (e)

|     |                                                                                                                           |                                                                                                                                                                            |
|-----|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\text{val}^\circ$                                                                                                        | the converse of $\text{val}$ , which reads digits as a number                                                                                                              |
| (b) | $([\text{wrap}, \text{snoc}]^\circ \text{Digit}^{++} (-\times \text{Digit}(\text{val})) [\text{embed}, \text{op}])^\circ$ | $\text{val}$ undoes $\text{wrap}$ or $\text{snoc}$ , reads the front with $\text{val}$ and combines by $\text{embed}$ or $\text{op}$ , so its converse runs that backwards |
| (c) | $(\text{wrap}^\circ \text{embed})^\circ \cup (\text{snoc}^\circ - \times \text{Digit}(\text{val}) \text{op})^\circ$       | undoing $\text{wrap}$ or $\text{snoc}$ first gives two cases: a decimal and its last digit read by $\text{val}$ then $\text{op}$ , or one digit read by $\text{embed}$     |
| (d) | $\text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val})^\circ \text{snoc}$                 | the converse of each case is its steps undone in reverse: undo $\text{op}$ , undo reading the front, then $\text{snoc}$ ; or undo $\text{embed}$ , then $\text{wrap}$      |
| (e) | $\text{embed}^\circ \text{wrap} \cup \text{op}^\circ - \times \text{Digit}(\text{val}^\circ) \text{snoc}$                 | undoing the read of the front while keeping the last digit is $\text{val}^\circ$ on the front alone                                                                        |

(6.1b)

$$\text{op}^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$$

$\text{op}^\circ$  splits a number into its quotient and remainder by 10

$$(\exists p. 0 < \pi_1 p \wedge \text{op}^\circ(m, p)) \iff 10 \leq m$$

$\text{op}^\circ$  gives a pair with a nonzero first component exactly at the numbers with two or more digits

$$(\exists d. \text{embed}^\circ(m, d)) \iff 0 < m \wedge m < 10$$

$\text{embed}^\circ$  gives a digit exactly at the one-digit numbers

## §6.2. Least fixed points

(6.2a)

$\varphi$  a **monotonic** mapping of the hom-set  $A \rightarrow B$  into itself:  $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$ .

$(\mu X : \varphi(X))$  the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$ .

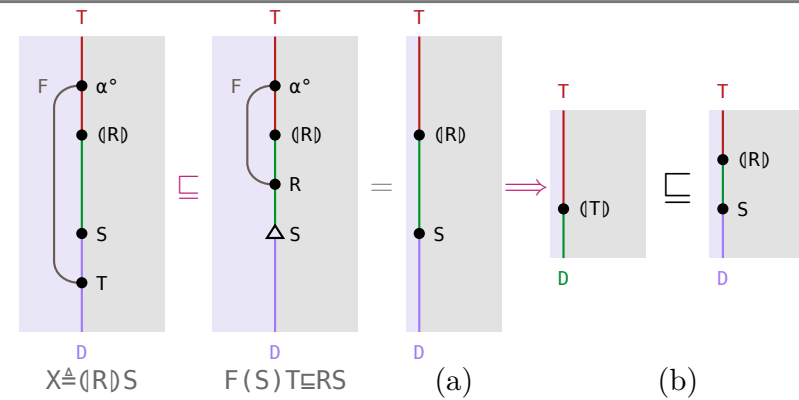
(6.2b)

| the law                                                                                    | what it says                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$                  | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §Section 6.3 and every chapter after it uses                                      |
| $\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$ | <b>Knaster–Tarski</b> : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |
| $\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$<br>(6.2)   | a relation $X$ that contains one unfolding of the fold's recursion at $X$ contains the fold                                                                                     |
| $X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$<br>(6.3)   | a relation $X$ contained in one unfolding of the fold's recursion at $X$ is contained in the fold                                                                               |

$$F(S) T \sqsubseteq R S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2c)

if  $S$  followed by  $R$  absorbs  $T$  after  $F(S)$ , the fold of  $T$  is below the fold of  $R$  followed by  $S$

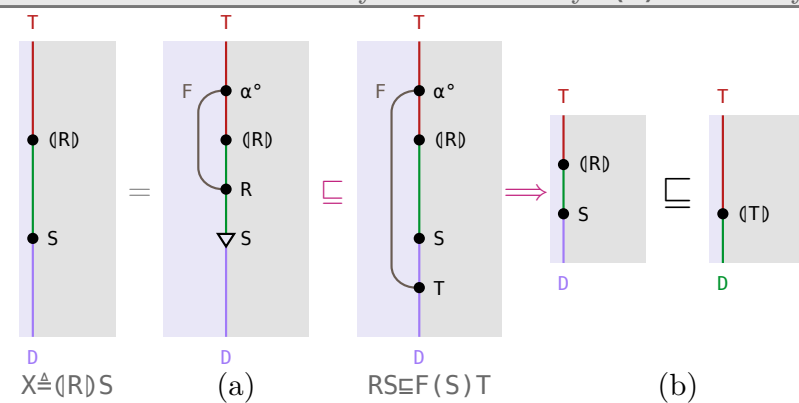


|     |                                                                                                                                       |                                                                                                                                                                                                    |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\langle R \rangle S$                                                                                                                 | $F(\langle R \rangle) R = \alpha(\langle R \rangle)$ , $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce                                                                                             |
| (b) | $\alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$ | $\langle R \rangle S$ is a prefix point of $\alpha^\circ F(-)T$ , and $\langle T \rangle$ is the least one — (6.2) $\alpha^\circ F(X) T \sqsubseteq X \Rightarrow \langle T \rangle \sqsubseteq X$ |

$$R S \sqsubseteq F(S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$$

(6.2d)

if  $R$  followed by  $S$  is absorbed by  $F(S)$  followed by  $T$ , the fold of  $R$  followed by  $S$  is below the fold of  $T$



|     |                                                                                                                                       |                                                                                                                                                                                                        |
|-----|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\alpha^\circ F(\langle R \rangle) R S$                                                                                               | $\alpha(\langle R \rangle) = F(\langle R \rangle) R$ , $\alpha^\circ \alpha = 1$ — $\triangleq$ reduce                                                                                                 |
| (b) | $\langle R \rangle S \sqsubseteq \alpha^\circ F(\langle R \rangle S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$ | $\langle R \rangle S$ is a postfix point of $\alpha^\circ F(-)T$ , and $\langle T \rangle$ is the greatest one — (6.3) $X \sqsubseteq \alpha^\circ F(X) T \Rightarrow X \sqsubseteq \langle T \rangle$ |

## §6.3. Hylomorphisms

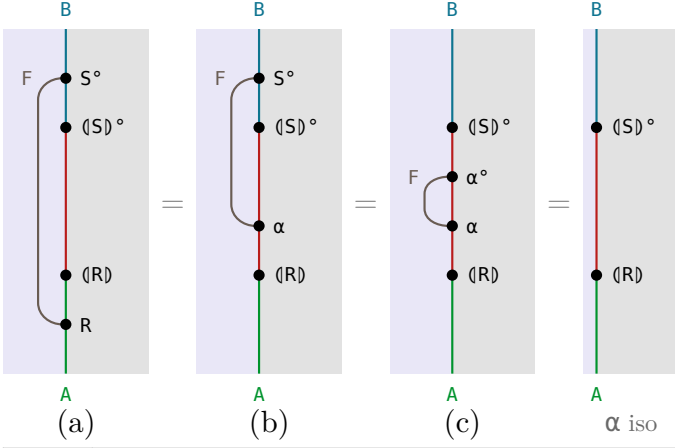
$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : \mathbb{S}^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$\mathbb{S}^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

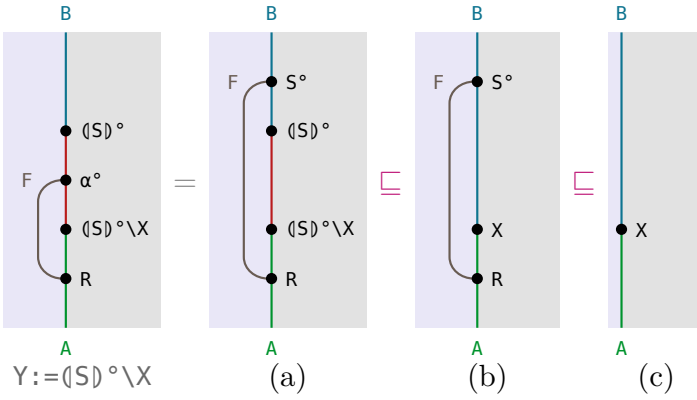
hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term  $\mathbb{S}^\circ$  represents the decomposition stage,  $F((\mathbb{S})^\circ(\mathbb{R}))$  the stage of solving the subproblems recursively, and  $R$  the recombination stage;  $R : FA \rightarrow A$ ,  $\mathbb{S} : FB \rightarrow B$ ,  $\alpha : FT \rightarrow T$  initial



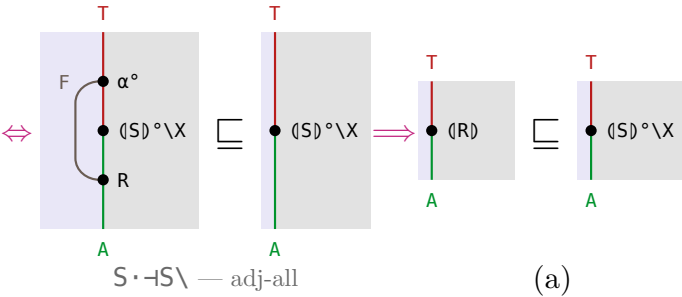
|     |                                                            |                                                                                                                   |
|-----|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| (a) | $\mathbb{S}^\circ F((\mathbb{S})^\circ)F((\mathbb{R}))R$   | the body at $(\mathbb{S})^\circ(\mathbb{R})$                                                                      |
| (b) | $\mathbb{S}^\circ F((\mathbb{S})^\circ)\alpha(\mathbb{R})$ | $F((\mathbb{R}))R = \alpha(\mathbb{R}) \triangleq \text{reduce}$                                                  |
| (c) | $(\mathbb{S})^\circ \alpha^\circ \alpha(\mathbb{R})$       | $(\mathbb{S})^\circ \alpha^\circ = \mathbb{S}^\circ F((\mathbb{S})^\circ) \triangleq \text{reduce, relator-laws}$ |

$$\mathbb{S}^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism  $(\mathbb{S})^\circ(\mathbb{R})$  is included in  $X$  if  $X$  satisfies the associated recursion inequation



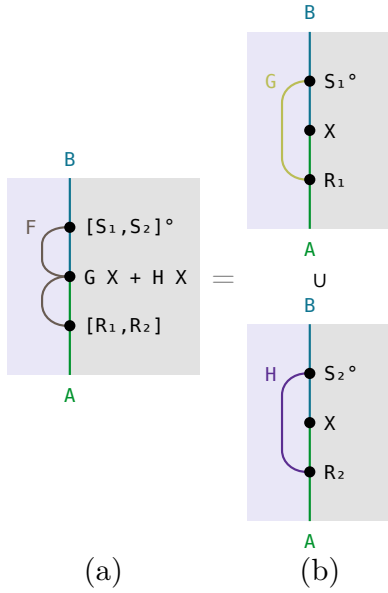
|     |                                                                           |                                                                                              |
|-----|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| (a) | $\mathbb{S}^\circ F((\mathbb{S})^\circ)((\mathbb{S})^\circ \setminus X)R$ | $(\mathbb{S})^\circ \alpha^\circ = \mathbb{S}^\circ F((\mathbb{S})^\circ)$                   |
| (b) | $\mathbb{S}^\circ F(X)R$                                                  | $(\mathbb{S})^\circ((\mathbb{S})^\circ \setminus X) \sqsubseteq X \triangleq \text{adj-all}$ |
| (c) | $X$                                                                       | $\mathbb{S}^\circ F(X)R \sqsubseteq X$                                                       |



|     |                                                                                                         |                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| (a) | $\mathbb{S}^\circ F(X)R \sqsubseteq X \implies (\mathbb{R}) \sqsubseteq (\mathbb{S})^\circ \setminus X$ | $(\mathbb{R}) = (\mu X : \alpha^\circ F(X)R) \triangleq \text{reduce, (6.2b), adj-all}$ |
|-----|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|

$$(\forall x. \forall y. \forall X. F(X)=G(X)+H(X)) \Rightarrow ([S_1, S_2]^\circ \llbracket [R_1, R_2] \rrbracket) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

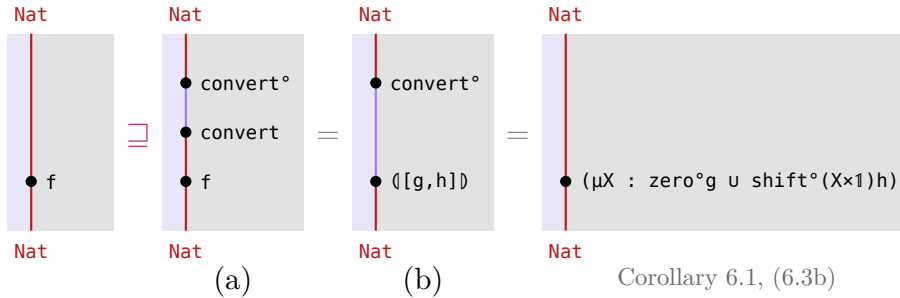


|     |                                              |                                                                                                                                                                                                                                                   |
|-----|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $[S_1, S_2]^\circ (G(X)+H(X)) [R_1, R_2]$    | the body of Theorem 6.2 at $F(X)=G(X)+H(X)$                                                                                                                                                                                                       |
| (b) | $S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$ | $[S, T] = (S+T) \nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$ ; the + stands for the codiagonals $\nabla^\circ, \nabla$ , which are not drawn<br>$\nabla^\circ (P+Q) \nabla = P \cup Q$ — coproduct |

## §6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1) h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1) h) \sqsubseteq f \quad (6.4a)$$

once **f** sends **zero** to **g** and turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



|     |                                                     |                                                                |
|-----|-----------------------------------------------------|----------------------------------------------------------------|
| (a) | $\text{convert}^\circ \text{convert } f$            | convert simple                                                 |
| (b) | $\text{convert}^\circ \llbracket [g, h] \rrbracket$ | fusion: $\text{zero } f = g, \text{shift } f = (f \times 1) h$ |

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes **a** to the power **b**

$\text{zero } \text{exp}(a) = \text{one}$

$a$  to the power  $0$  is  $1$

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

$a$  to the power  $2n+d$  is  $\text{op}(a)$  of  $a$  to the power  $n$  and  $d$

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves **a** at each step computes the remainder of **a** by **b**

$\text{zero } \text{mod}(b) = \text{zero}$

$0 \text{ mod } b$  is  $0$

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

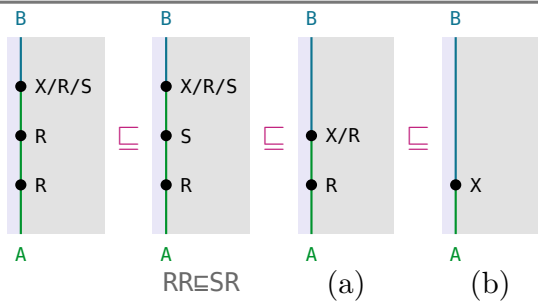
$(2a+d) \text{ mod } b$  is  $\text{op}(b)$  of  $a \text{ mod } b$  and  $d$

## §6.5. Unique fixed points

$$\text{inductive}(S) \wedge R \in SR \Rightarrow \text{inductive}(R)$$

(6.5a)

$R$  is inductive when  $R \in SR$  for an inductive  $S$



|     |          |                 |
|-----|----------|-----------------|
| (a) | $(X/R)R$ | division by $S$ |
| (b) | $X$      | division by $R$ |

$$\text{member} = \emptyset$$

(6.5b)

a constant relator records no elements, so its membership is empty

$$\text{member} = [\text{member}, \text{member}]$$

a member of  $F+G$  is a member of whichever summand the value is in

$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of  $F \times G$  is a member of either component

$$\text{inductive}(\text{Smember}) \wedge X = SF(X)R \wedge Y = SF(Y)R \Rightarrow X = Y$$

(6.5c)

when  $S$  member is inductive, two solutions  $X, Y$  of  $X = SF(X)R$  are equal

$$\text{inductive}(\text{Smember}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge SF(X)R \in X \Rightarrow \text{Entire}(X)$$

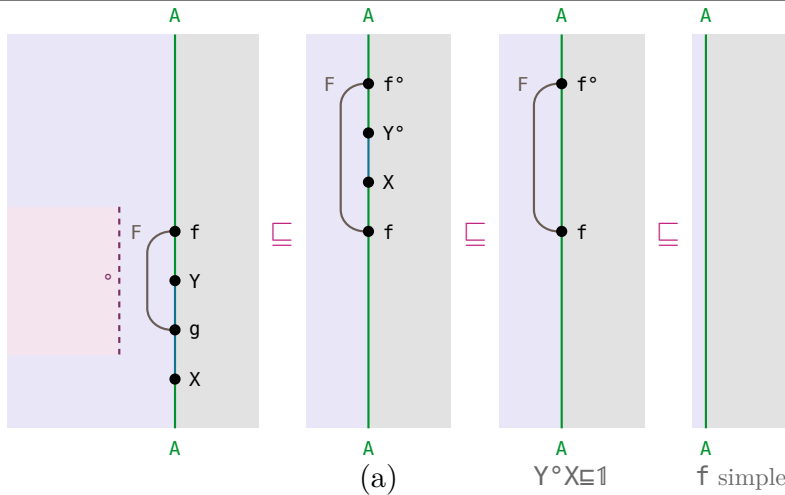
(6.5d)

when  $S$  member is inductive and  $S, R$  are entire, a pre-fixed point of  $X \mapsto SF(X)R$  is entire

$$\text{inductive}(\text{gmember}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = gF(X)f \Rightarrow \text{Map}(X)$$

(6.5e)

when  $g$  member is inductive and  $f, g$  are maps, the solution of  $X = gF(X)f$  is a map

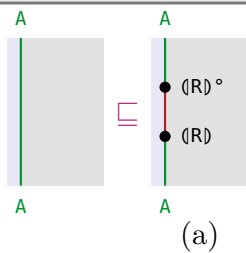


|     |                          |                                 |
|-----|--------------------------|---------------------------------|
| (a) | $f^\circ F(Y^\circ X) f$ | $X = gF(X) f, g \text{ simple}$ |
|-----|--------------------------|---------------------------------|

$$\text{inductive}(R^\circ \text{member}) \wedge 1 \in R^\circ R \Rightarrow 1 \in (R)^\circ (R)$$

(6.5f)

when  $R^\circ \text{member}$  is inductive and  $R$  is surjective, the fold  $(R)$  is surjective

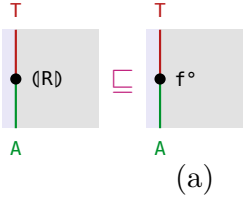


|     |                 |                         |
|-----|-----------------|-------------------------|
| (a) | $(R)^\circ (R)$ | (6.5d) at $S = R^\circ$ |
|-----|-----------------|-------------------------|

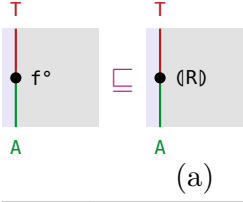
$$\text{Map}(f) \wedge R \sqsubseteq F(f) \alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = \langle R \rangle$$

(6.5g)

a map  $f$  with  $R \sqsubseteq F(f) \alpha$ , for a surjective  $R$ , has the fold  $\langle R \rangle$  as its converse



(a)

 $f^\circ$ shunting  $f$ , fusion,  $R \sqsubseteq F(f) \alpha$ 

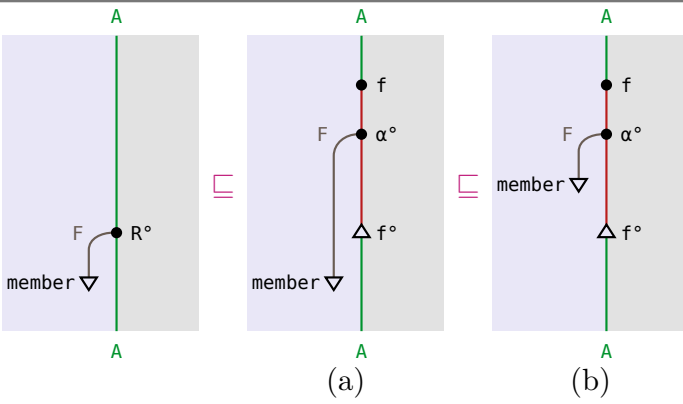
(a)

 $\langle R \rangle$ (6.5f) and  $\langle R \rangle f \sqsubseteq 1$ 

$$\text{Map}(f) \wedge R \sqsubseteq F(f) \alpha \Rightarrow R^\circ \text{member} \sqsubseteq f \alpha^\circ \text{member } f^\circ$$

(6.5h)

$R^\circ \text{member}$  is below  $\alpha^\circ \text{member}$  conjugated by  $f$ , so it is inductive when  $\alpha^\circ \text{member}$  is



(a)

 $f \alpha^\circ F(f)^\circ \text{member}$  $R \sqsubseteq F(f) \alpha$ , shunting

(b)

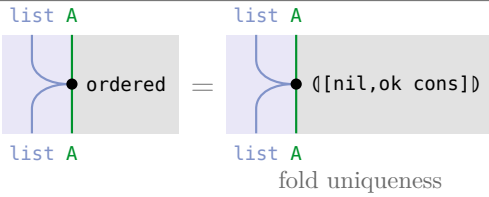
 $f \alpha^\circ \text{member } f^\circ$  $\text{member}$  lax natural

## §6.6. Sorting by selection

$$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$$

(6.6a)

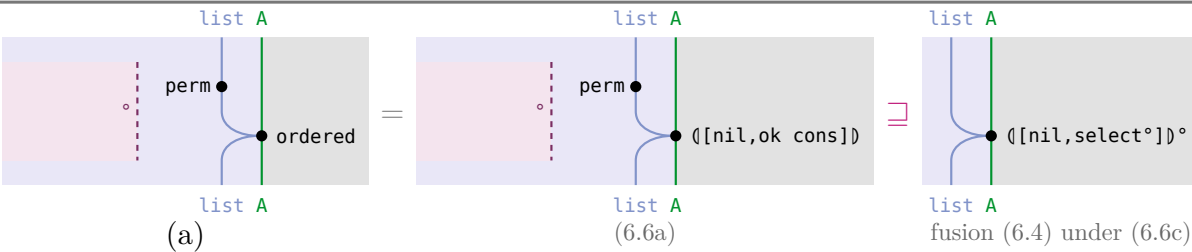
a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it



$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow \langle [\text{nil}, \text{select}^\circ] \rangle^\circ \sqsubseteq \text{perm ordered}$$

(6.6b)

every output of unfolding the input by **select** is a sorted permutation of it

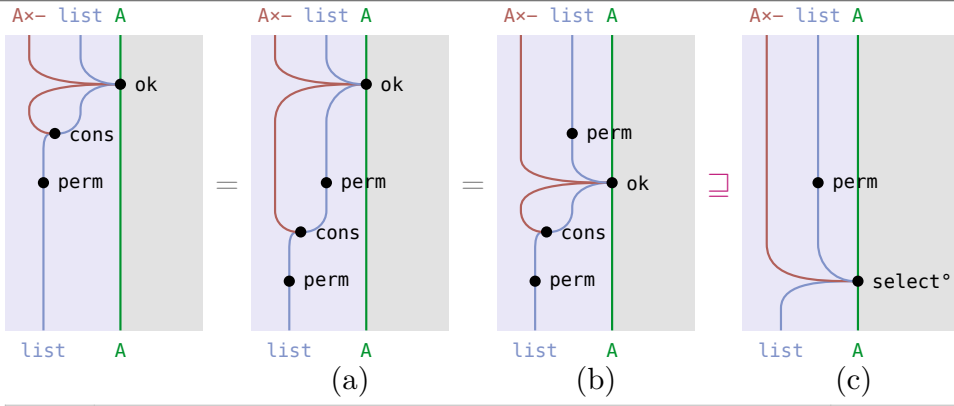


(a)

 $(\text{ordered perm})^\circ$  $\text{perm}^\circ = \text{perm}$ ,  $\text{ordered}^\circ = \text{ordered}$

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \sqsubseteq \text{ok cons perm} \quad (6.6c)$$

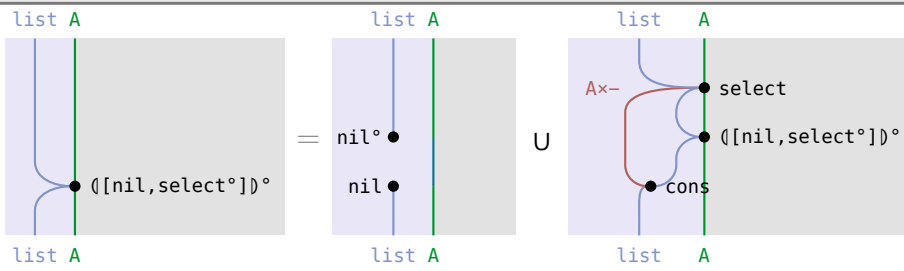
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



|     |                                                      |                                                                               |
|-----|------------------------------------------------------|-------------------------------------------------------------------------------|
| (a) | $\text{ok } (1 \times \text{perm}) \text{cons perm}$ | $\text{cons perm} = (1 \times \text{perm}) \text{cons perm}$                  |
| (b) | $(1 \times \text{perm}) \text{ok cons perm}$         | Ex 6.22, $\text{ok}(1 \times \text{perm}) = (1 \times \text{perm}) \text{ok}$ |
| (c) | $(1 \times \text{perm}) \text{select}^\circ$         | $\text{select}^\circ \sqsubseteq \text{ok cons perm}$                         |

$$([nil, \text{select}^\circ])^\circ = nil^\circ nil \cup \text{select } (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

selection sort returns  $[]$  on  $[]$ , and otherwise selects  $(a, y)$ , sorts  $y$  and conses  $a$  back on

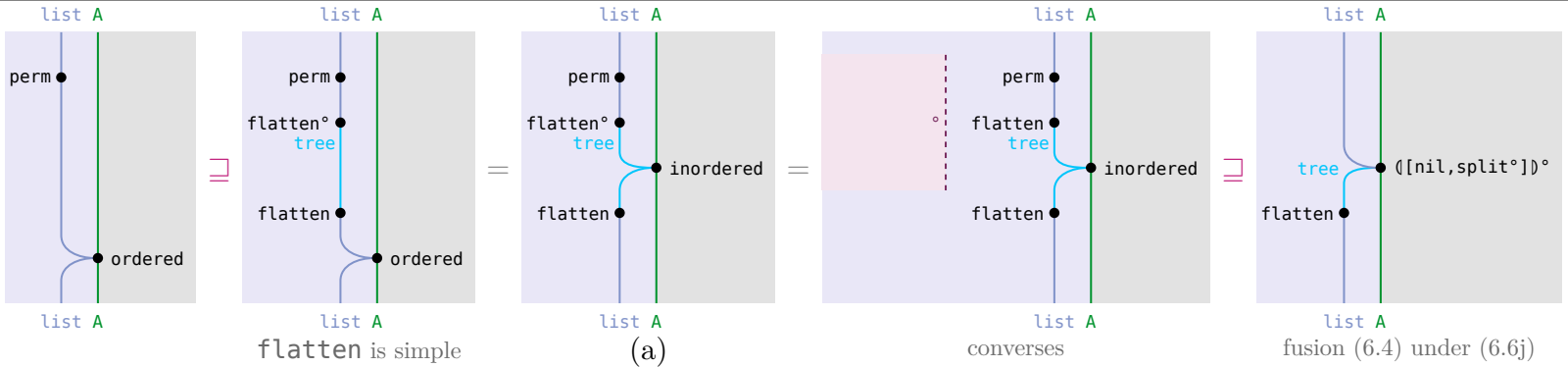


unfolding the converse of a fold

$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \wedge \text{split}^\circ \sqsubseteq \text{check}' \text{ join perm} \Rightarrow \quad (6.6e)$$

$$([nil, \text{split}^\circ])^\circ \text{flatten} \sqsubseteq \text{perm ordered}$$

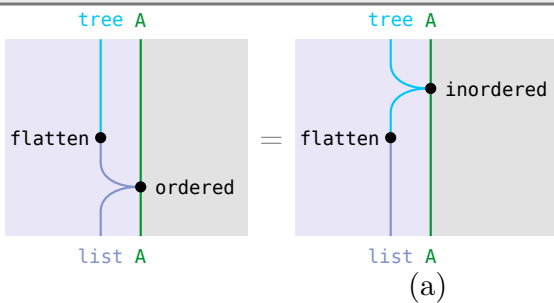
every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input



|     |                                                      |                                                     |
|-----|------------------------------------------------------|-----------------------------------------------------|
| (a) | $\text{perm flatten}^\circ \text{inordered flatten}$ | $\text{flatten ordered} = \text{inordered flatten}$ |
|-----|------------------------------------------------------|-----------------------------------------------------|

$$(\forall a. \forall b. \forall c. R(a, b) \Rightarrow R(b, c) \Rightarrow R(a, c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten} \quad (6.6f)$$

flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening

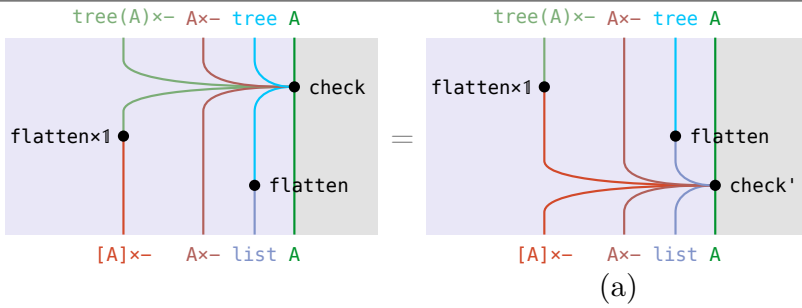


|     |                            |                                                                       |
|-----|----------------------------|-----------------------------------------------------------------------|
| (a) | $\text{inordered flatten}$ | R transitive: a flattened tree is ordered iff every node passes check |
|-----|----------------------------|-----------------------------------------------------------------------|

**check (flatten×(1×flatten))=(flatten×(1×flatten))check'**

(6.6g)

testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



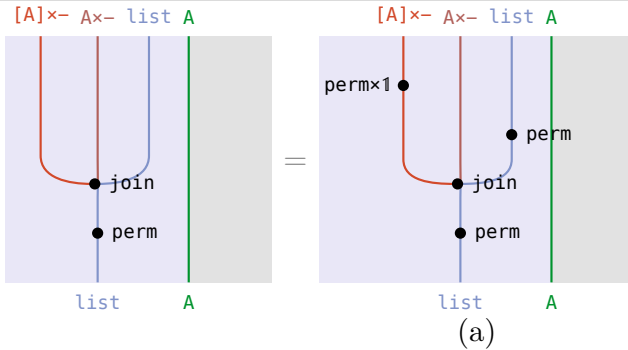
(a)  $(\text{flatten} \times (1 \times \text{flatten})) \text{check}'$

$b \text{ intree } x \iff b \text{ inlist } \text{flatten}(x)$

**join perm=(perm×(1×perm))join perm**

(6.6h)

permuting the list  $x ++ [a] ++ y$  is the same as first permuting  $x$  and  $y$  and then permuting the joined list



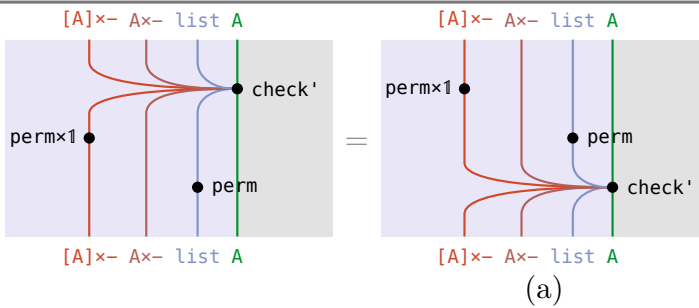
(a)  $(\text{perm} \times (1 \times \text{perm})) \text{join perm}$

$\text{perm}$  is transitive and closed under  $++$

**check' (perm×(1×perm))=(perm×(1×perm))check'**

(6.6i)

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

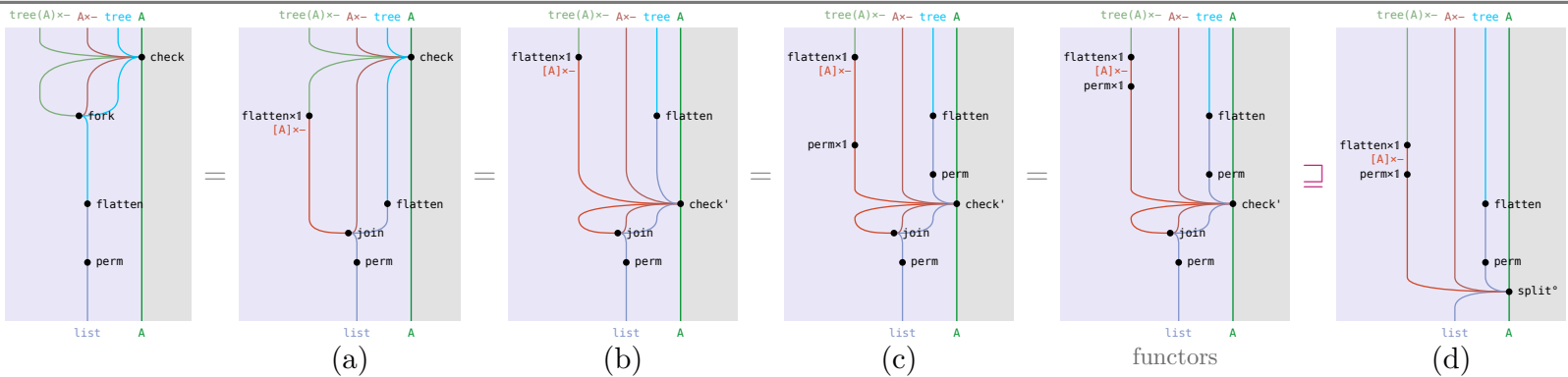


(a)  $(\text{perm} \times (1 \times \text{perm})) \text{check}'$

$b \text{ inlist } x \iff b \text{ inlist } \text{perm}(x)$

**split° ⊆ check' join perm ⇒ ((flatten perm)×(1×(flatten perm)))split° ⊆ check fork flatten perm** (6.6j)

undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted

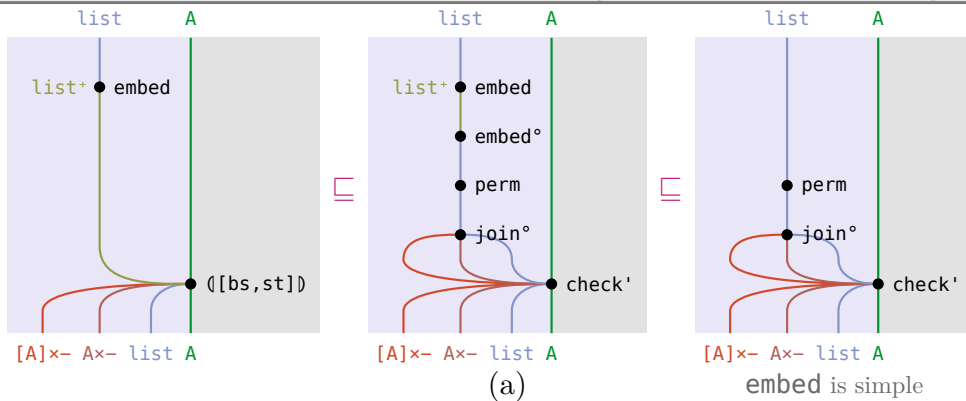


|     |                                                                                                                                |                                                                                                                 |
|-----|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| (a) | $\text{check} (\text{flatten} \times (1 \times \text{flatten})) \text{join perm}$                                              | $\text{fork flatten} = F(\text{flatten}) \text{join}$                                                           |
| (b) | $(\text{flatten} \times (1 \times \text{flatten})) \text{check}' \text{join perm}$                                             | $\text{check } F(\text{flatten}) = F(\text{flatten}) \text{check}'$                                             |
| (c) | $(\text{flatten} \times (1 \times \text{flatten})) (\text{perm} \times (1 \times \text{perm})) \text{check}' \text{join perm}$ | $\text{join perm} = F(\text{perm}) \text{join perm, check'}$<br>$F(\text{perm}) = F(\text{perm}) \text{check}'$ |
| (d) | $((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^\circ$                                           | $\text{split}^\circ \subseteq \text{check}' \text{join perm}$                                                   |

$$\text{bs}\text{Ewrap perm join}^\circ\text{check}' \wedge (1 \times (\text{perm join}^\circ\text{check}')) \text{stEcons perm join}^\circ\text{check}' \Rightarrow \text{embed}([bs, st]) \text{D} \quad (6.6k)$$

**perm join<sup>°</sup>check'**

turning a list into a non-empty list and folding it with any  $[bs, st]$  meeting the two conditions below gives only **checked triples**  $(x, a, y)$  whose join  $x \# [a] \# y$  is a permutation of the list



(a)

embed is simple

(a)

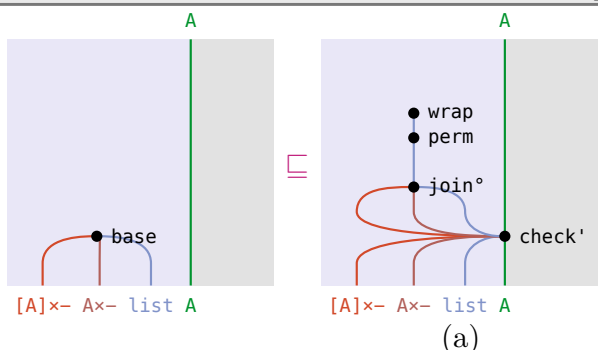
embed embed<sup>°</sup>perm join<sup>°</sup>check'

(6.6l) and (6.6m), induction on the non-empty list

**baseEwrap perm join<sup>°</sup>check'**

(6.6l)

**base(a)** is a **checked triple** whose join is a permutation of the one-element list  $[a]$



(a)

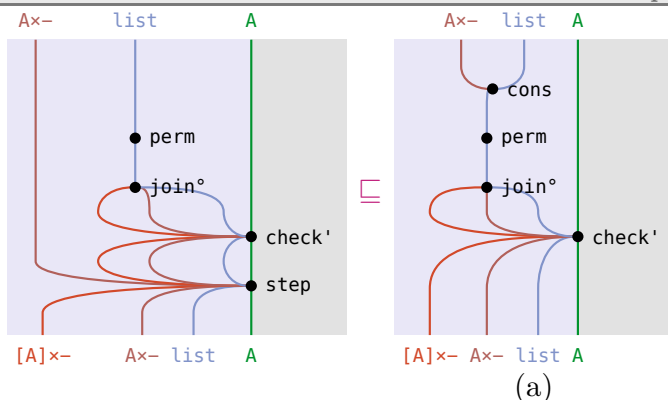
(a)

wrap perm join<sup>°</sup>check'

[] # [a] # [] = [a], and check' holds on empty lists

$$(\forall a. \forall b. \text{leb}(a, b) = \text{true} \Rightarrow R(a, b)) \wedge (\forall a. \forall b. \text{leb}(a, b) = \text{false} \Rightarrow R(b, a)) \Rightarrow (1 \times (\text{perm join}^\circ\text{check}')) \text{stepEcons perm join}^\circ\text{check}' \quad (6.6m)$$

if  $(x, b, y)$  is a **checked triple** for a permutation of  $l$ , then **step(a, (x, b, y))** is a **checked triple** for a permutation of  $[a] \# l$



(a)

(a)

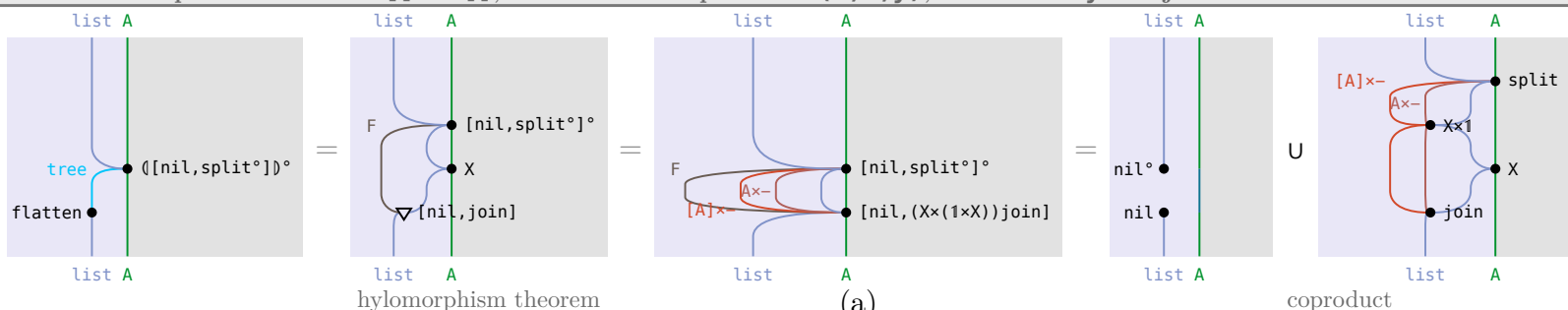
cons perm join<sup>°</sup>check'aRb puts **a** in front of **x**; otherwise **bRa** and **a** goes in front of **y**

$$([nil, \text{split}^\circ])^\circ \text{flatten} = nil^\circ nil \cup \text{split}$$

(6.6n)

$$((([nil, \text{split}^\circ])^\circ \text{flatten}) \times (1 \times ([nil, \text{split}^\circ])^\circ \text{flatten})) \text{join}$$

quicksort returns  $[]$  on  $[]$ , and otherwise splits into  $(x, a, y)$ , sorts  $x$  and  $y$  and joins them around  $a$



hylomorphism theorem

(a)

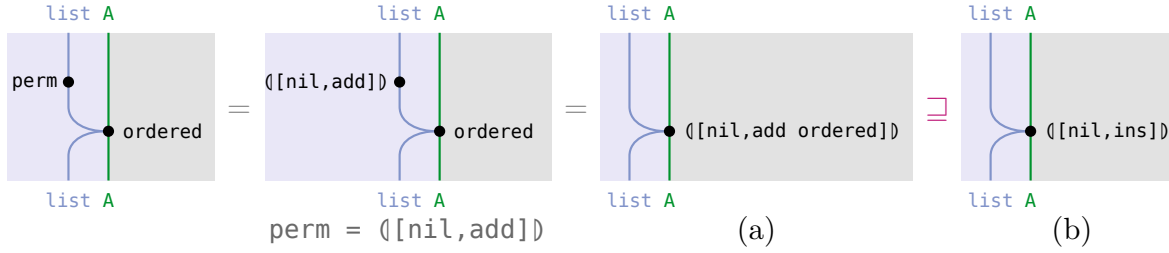
coproduct

(a)

 $[nil, \text{split}^\circ]^\circ [nil, (X \times (1 \times X)) \text{join}]$  $F(X) [nil, \text{join}] = [nil, (X \times 1 \times X) \text{join}]$

$$(1 \times \text{ordered}) \text{ins} \sqsubseteq \text{add ordered} \Rightarrow ([\text{nil}, \text{ins}]) \sqsubseteq \text{perm ordered} \quad (6.6o)$$

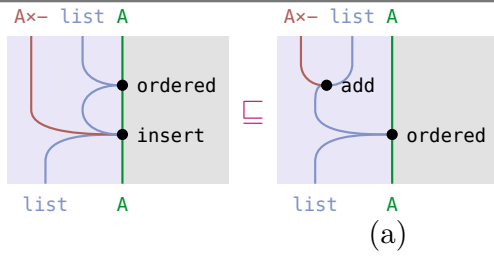
for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations



|     |                                        |                                                                             |
|-----|----------------------------------------|-----------------------------------------------------------------------------|
| (a) | $([ \text{nil}, \text{add ordered} ])$ | fusion, $\text{add ordered} = (1 \times \text{ordered}) \text{add ordered}$ |
| (b) | $([ \text{nil}, \text{ins} ])$         | $(1 \times \text{ordered}) \text{ins} \sqsubseteq \text{add ordered}$       |

$$(\forall a. \forall b. \text{leb}(a,b)=\text{true} \Rightarrow R(a,b)) \wedge (\forall a. \forall b. \text{leb}(a,b)=\text{false} \Rightarrow R(b,a)) \wedge (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow (1 \times \text{ordered}) \text{insert} \sqsubseteq \text{add ordered} \quad (6.6p)$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



|     |                      |                                                                        |
|-----|----------------------|------------------------------------------------------------------------|
| (a) | $\text{add ordered}$ | $\text{insert}$ splices <b>a</b> in, and <b>R</b> transitive and total |
|-----|----------------------|------------------------------------------------------------------------|

## §6.7. Closure

$$R^* = (\mu X : 1 \cup XR) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R\***

$$1 \sqsubseteq X \wedge XX \sqsubseteq X \Rightarrow RX \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \quad (6.7b)$$

for a preorder **X**, **X** contains **R** exactly when it contains **R\***

$$1 \sqsubseteq R^* \quad (6.7c)$$

**R\*** is reflexive

|               |     |               |                      |
|---------------|-----|---------------|----------------------|
| $\sqsubseteq$ | (a) | $1$           |                      |
| $\sqsubseteq$ | (b) | $1 \cup RR^*$ | union                |
| $=$           | (c) | $R^*$         | fixed point, Thm 6.1 |

$$R \sqsubseteq R^* \quad (6.7d)$$

**R\*** contains **R**

|               |     |        |                                      |
|---------------|-----|--------|--------------------------------------|
| $=$           | (a) | $R$    |                                      |
| $\sqsubseteq$ | (b) | $R$    | identity                             |
| $\sqsubseteq$ | (c) | $RR^*$ | $1 \sqsubseteq R^* \text{ — (6.7c)}$ |
| $\sqsubseteq$ | (d) | $R^*$  | fixed point                          |

$$R^* R^* \sqsubseteq R^* \quad (6.7e)$$

**R\*** is transitive, because **R\*/R\*** is a prefixed point of  $X \mapsto 1 \cup RX$

|               |     |                          |                                                                         |
|---------------|-----|--------------------------|-------------------------------------------------------------------------|
| $=$           | (a) | $(1 \cup R(R^*/R^*))R^*$ |                                                                         |
| $\sqsubseteq$ | (b) | $R^* \cup R(R^*/R^*)R^*$ | $(P \cup Q)T = PT \cup QT \text{ — composition distributes over } \cup$ |
| $\sqsubseteq$ | (c) | $R^* \cup RR^*$          | $(S/S)S \sqsubseteq S$                                                  |
| $\sqsubseteq$ | (d) | $R^*$                    | $RR^* \sqsubseteq R^* \text{ — (6.7d)}$                                 |

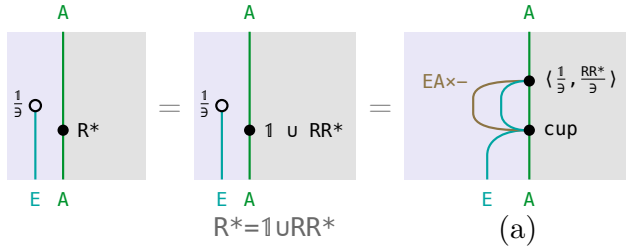
$$1 \sqsubseteq X \wedge XX \sqsubseteq X \wedge RX \sqsubseteq X \Rightarrow R^* \sqsubseteq X \quad (6.7f)$$

a reflexive transitive **X** containing **R** is a prefixed point of  $X \mapsto 1 \cup RX$ , so it contains **R\***

|               |     |             |                                    |
|---------------|-----|-------------|------------------------------------|
| $\sqsubseteq$ | (a) | $1 \cup RX$ |                                    |
| $\sqsubseteq$ | (b) | $X \cup XX$ | $1 \sqsubseteq X, R \sqsubseteq X$ |
| $\sqsubseteq$ | (c) | $X$         | $XX \sqsubseteq X$                 |

$$\frac{R^*}{S} = (\frac{1}{S}, \frac{RR^*}{S}) \text{ cup} \quad (6.7g)$$

the set of  $R^*$ -successors of  $a$  is  $a$  itself joined with the  $R^*$ -successors of its  $R$ -successors



(a)

$$(\frac{1}{S}, \frac{RR^*}{S}) \text{ cup}$$

$$\Lambda(R \cup S) = (\Lambda(R), \Lambda(S)) \text{ cup}$$

$$R - \emptyset = R \quad (6.7h)$$

taking nothing away leaves  $R$

$$R \cup (S - R) = R \cup S$$

beside  $R$ , only the part of  $S$  outside  $R$  adds anything

$$R - (S \cup T) = R - S - T$$

taking away  $S$  and then  $T$  takes away  $S \cup T$

$$(R \cup S) - T = (R - T) \cup (S - T)$$

taking  $T$  away from a union takes it away from each part

$$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y))) \quad (6.7i)$$

the least fixed point of  $\varphi$  after  $\psi$  is  $\varphi$  applied to the least fixed point of  $\psi$  after  $\varphi$

$$SR^* = (\mu X : S \cup XR) \quad (6.7j)$$

$S$  followed by any number of  $R$  steps is the least  $X$  containing  $S$  and closed under a further  $R$  step

$$R^*S = (\mu X : S \cup RX)$$

any number of  $R$  steps followed by  $S$  is the least  $X$  containing  $S$  and closed under an  $R$  step in front

$$\theta(\emptyset, S) = SR^* \quad (6.7k)$$

started with nothing found and  $S$  to explore,  $\theta$  computes  $SR^*$

|   |     |                                                    |                              |
|---|-----|----------------------------------------------------|------------------------------|
|   | (a) | $\theta(\emptyset, S)$                             |                              |
| = | (b) | $\emptyset \cup (\mu X : S \cup (XR - \emptyset))$ | definition of $\theta$       |
| = | (c) | $(\mu X : S \cup XR)$                              | $R - \emptyset = R$ — (6.7h) |
| = | (d) | $SR^*$                                             | (6.7j)                       |

$$\theta(P, \emptyset) = P \quad (6.7l)$$

with nothing left to explore,  $\theta$  returns what it has found

|   |     |                                            |                                                                |
|---|-----|--------------------------------------------|----------------------------------------------------------------|
|   | (a) | $\theta(P, \emptyset)$                     |                                                                |
| = | (b) | $P \cup (\mu X : \emptyset \cup (XR - P))$ | definition of $\theta$                                         |
| = | (c) | $P \cup \emptyset$                         | $\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$ |
| = | (d) | $P$                                        | $P \cup \emptyset = P$                                         |